INCH-POUND

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MILITARY SPECIFICATION

PORK AND BEANS, THERMOSTABILIZED, TRAY PACK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers pork and beans, thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard

MILITARY

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification for
MIL-C-44340 - Can, Tray Pack

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies are available from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

U.S. DEPARIMENT OF AGRICULTURE (USDA)

Regulations Governing Inspection and Certification of Processed Fruits and Vegetables, Processed Products Thereof, and Certain Other Processed Food Products (7 CFR Part 52)

U.S. Standards for Grades of Tomato Paste (7 CFR Part 52)

U.S. Standards for Grades of Sugarcane Molasses (7 CFR Part 52)

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(Copies are available from the Chief, Processed Products Branch, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 0709, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

U.S. Standards for Beans (7 CFR Part 68)

(Copies are available from the Federal Grain Inspection Service, APHIS, Printing and Distribution Section, 6505 Belcrest Road, Room G-100 Federal Building, Hyattsville, MD 20782.)

U.S. Standards for Condition of Food Containers

(Copies are available from the Chairperson, Condition of Container Committee, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2506, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS), U.S. FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, 650 California Street, Suite 800, San Francisco, CA 94108.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3330 - Peel Adhesion of Pressure-sensitive Tape

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AOAC INTERNATIONAL

Official Methods of Analysis of the AOAC

(Application for copies should be addressed to the AOAC International, 2200 Wilson Boulevard, Suite 400, Arlington, VA 22201-3301.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, NW, Washington, DC 20418.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Pea beans, dry (navy beans). Pea beans shall be dry, mature beans (the type as grown in the Great Lakes region known as navy beans). The dry beans shall be U.S. No. 1 or better of the U.S. Standards for Beans, shall have a moisture content of not more than 18 percent, and shall be of the latest season's crop.

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preclude the prior freezing of the product the freeze dryer as detailed in 4.4.1

3.2.2 Bacon (cured and smoked), either presliced or sliced, ends and pieces. Sliced, cured and smoked bacon ends and pieces or presliced bacon shall be used. The bacon shall meet the size requirements specified in 3.3.2. Nonvacuum-packaged bacon shall be held at an internal temperature of 28°F to 40°F and shall be used in the finished product preparation within ten days after removal from the smokehouse. Vacuum-packaged bacon may be held at an internal temperature of 28°F to 40°F for not more than 30 days. Frozen bacon packaged in USDA approved food packaging material may be held at an internal temperature of 0°F or lower for not more than 60 days.

3.2.3 Water. Water used for ice making, formulation, blanching, washing, and processing shall conform to the National Primary Drinking Water Regulations.

3.2.4 Tomato paste. Tomato paste shall be U.S. Grade A of the U.S. Standards for Grades of Tomato Paste.

3.2.5 Molasses, sugarcane. Sugarcane molasses shall be what is commonly called dark molasses. The chemical analysis shall be as follows:

Brix solids	-	Not less than 77.0 percent
Total sugar	-	Not less than 58.0 percent
Ash	-	Not more than 9.0 percent
Total sulfites	-	Not more than 250 ppm

The certificate of conformance (COC) shall state the chemical analysis. Alternatively, the product shall be accompanied by a USDA Certificate certifying the molasses as U.S. Grade C or better of the U.S. Standards for Grades of Sugarcane molasses.

3.2.6 Sugar, brown, dark. Brown sugar shall be cane or beet sugar and shall be dark brown in color.

3.2.7 Onion powder. Onion powder shall be Fancy Grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic products.

3.2.8 Salt. Salt shall be non-iodized, white, refined, sodium chloride with or without anticaking agents and shall comply with the purity standards for sodium chloride of the Food Chemicals Codex.

3.2.9 Starch, modified, high opacity. The high opacity modified starch shall be white, odorless, finely pulverized, modified maize food starch for use in thermostabilized foods. The starch shall demonstrate initial viscosity development in the temperature range of 140°F to 170°F and be fully hydrated at common retort temperatures. The starch shall resist breakdown at low pH and under shear stress. The cooked slurry prepared from the starch shall be bland with essentially no cereal or starch taste.

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3.2.10 Mustard flour. Mustard flour shall be a bright yellow powder from a blend derived from the endosperm of the seed of Brassica hirta and Brassica juncea. Mustard flour shall contain not less than 0.4 mL of volatile oil per 100 grams of mustard flour and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 60 sieve.

3.2.11 Garlic powder. Garlic powder shall be Fancy Grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.12 Pepper, white, ground. Ground white pepper shall be derived from the dried mature berries of Piper nigrum L. from which the outer covering or the outer and inner coverings have been removed. The ground pepper shall have a characteristic, penetrating odor, a hot, biting, pungent flavor, and a light color. The ground pepper shall contain not less than 1.0 mL of volatile oil per 100 grams of ground white pepper and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 40 sieve.

3.2.13 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements of this specification. The containers used for the spice and seasoning mixture shall be labeled with each ingredient and the percentage of each ingredient in the mixture. The ingredients shall be in the same proportions as specified in the sauce formula (see 3.3.3).

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Bean preparation. The beans shall be thoroughly cleaned to remove dirt, stones, loose skins, and any other extraneous material, and sorted to remove those beans which are spotted, discolored, or affected by pathological or insect injury. The beans shall be thoroughly washed and soaked to a moisture content of 54 percent to 60 percent (approximately eight to sixteen hours in cold water). The soaked beans shall be thoroughly drained. The beans shall be blanched sufficiently to prevent discoloration and to remove excess air. The blanched beans shall be immediately cooled to the initial temperature of the cooling water and thoroughly drained. The cooled, drained beans shall be handled in a manner to prevent discoloration and filled into the tray pack cans within four hours after blanching.

3.3.2 Bacon preparation. The bacon shall be cut into 3/4 inch to 1-1/4 inch pieces.

3.3.3 Preparation of the sauce. The sauce shall be formulated and prepared as follows:

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Ingredient	Percent by weight
Water	70.70
Tomato paste (28% natural tomato soluble solids) 1/	12.64
Molasses, sugarcane	6.38
Sugar, brown, dark	6.40
Onion powder	1.25
Salt 2/	2.03
Starch, modified, high opacity	0.50
Mustard flour	0.06
Garlic powder	0.03
Pepper, white, ground	0.01

1/ If the specified tomato solids are not available, the tomato solids that are used must be calculated to bring the solids level to that specified and the water shall be adjusted accordingly.

2/ The total amount of salt in the sauce formula shall be adjusted, as necessary, to produce a product that complies with the finished product salt requirements (see 3.6).

NOTE: The following sauce preparation procedures were used in the development of this product. Alternate procedures may be used provided finished product requirements are met. (When alternate procedures are used, the time and temperature requirements specified for the prepared sauce are still applicable.)

The sauce shall be prepared as follows:

a. A thin slurry shall be made with the starch and part of the water.

b. The balance of the water and remaining ingredients shall be mixed together and heated to 140°F to 180°F to form an emulsion.

c. The starch slurry shall be added to the emulsion to form the sauce.

d. The sauce shall be heated to 180°F to 190°F and held for approximately five minutes in this temperature range.

e. The volume of the sauce shall be adjusted with water to compensate for evaporation loss during heating and holding.

f. The sauce shall be maintained in the temperature range of 150°F to 180°F and filled into tray pack cans within four hours after preparation.

3.3.4 Product preparation. The beans, bacon, and sauce shall be combined in the following proportions using a multistage filling operation. The percent by weight of the sauce and beans may be adjusted, if necessary, to ensure compliance with the finished product drained weight requirements.

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Ingredient	Percent by weight
Sauce	49.00
Beans, soaked, blanched	48.10
Bacon, cut	2.90

3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with product such as to conform to the finished product requirements and to the following requirements:

a. Immediately after filling, each can shall be sealed in accordance with the can manufacturer's guidelines/requirements and 21 CFR, Part 113, Subpart D, or CFR 9, Part 318, Subpart G, as applicable (see 4.5.5), and under a vacuum established by a processing authority and specified in the scheduled process so as to ensure compliance with the finished product requirement (see 3.6).

b. The filled and sealed tray pack cans shall be in the retort process within two hours after sealing.

3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting until a sterilization value (F_o) of not less than 6 has been achieved. The filled and sealed thermoprocessed tray pack cans shall show no evidence of can swelling when tested for commercial sterility as specified in 4.5.3.4.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, musty, or moldy.

c. There shall be no color foreign to the product.

d. No individual can shall contain less than 104 ounces of product.

e. The average net weight shall be not less than 106 ounces.

f. Sauce consistency shall be smooth, without lumps, and shall not be excessively thick or thin.

g. Sauce shall be a uniform dark reddish brown color.

h. Beans shall have a uniform brown color that is typical of canned navy pea beans packed in tomato sauce.

i. Washed drained weight of the product in any individual tray pack can shall be not less than 62.0 ounces nor more than 72.0 ounces.

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j. Not more than 5 percent, by weight, of the washed drained weight of product shall contain broken beans, mashed beans, or beans with loose skins.

k. The product shall show no evidence of excessive heating (materially darkened or scorched).

l. Texture of the beans shall be tender but not mushy.

m. Bean skins shall not be tough.

n. The salt content of the finished product shall be not less than 0.6 percent nor greater than 1.5 percent.

o. Filled, sealed, and retorted cans shall show evidence of proper vacuum as determined by concavity of the can lid (see 4.5.6).

3.6.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Packing or Holding Human Food," and the plant sanitation requirements of the appropriate Government inspection agency.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection and supervision necessary to ensure compliance with the requirements of this specification.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

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[illegible] temperature of 28 F to 30 F after removal from the cooking

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material examination. In accordance with 4.1, components and materials shall be examined in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the methods described in the Official Methods of Analysis of the AOAC and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to preparation, processing, can interior coating, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Tray pack inspection. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.5.3.1 Net weight examination. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 1 ounce. If the average net weight is less than 106 ounces, or if the net weight of any individual can is less than 104 ounces, the lot shall be rejected.

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4.5.3.2 Double sampling plan for product examination. The finished product shall be examined for the defects listed in table I utilizing the double sampling plans indicated in MIL-STD-105. The lot size shall be expressed in tray pack cans. The sample unit shall be one filled and sealed tray pack can. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0 for major defects. The sample cans shall be held for a minimum of 48 hours, but not longer than 72 hours in a temperature range of 65°F to 75°F after completing the thermoprocessing procedure. The sample cans shall be heated in accordance with the heating instructions on the can label.

TABLE I. Product defects 1/ 2/ 3/

Category	Defect
Major	
101	Sauce consistency is lumpy or is excessively thick or thin
102	Color of the sauce not a uniform dark reddish brown
103	Beans not uniformly brown in color typical of canned navy pea beans packed in tomato sauce
104	Washed drained weight of the product in any individual tray pack can is less than 62.0 ounces or more than 72.0 ounces 4/
105	More than 5 percent, by weight, of the washed drained weight of product contains broken or mashed beans or beans with loose skins
106	Product shows evidence of excessive heating (materially darkened or scorched)
107	Texture of the beans is mushy or hard
108	Bean skins are tough

1/ The presence of foreign material (for example, dirt, insect parts, hair, wood, glass, metal), foreign odor or flavor (for example, burnt, scorched, stale, sour, rancid, musty, moldy), or foreign color shall be cause for rejection of the lot.

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2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot (see 3.6.1).

3/ Stones shall be defined as extraneous material. If one stone is found in two or more cans the lot shall be rejected. If more than one stone is found in any can the lot shall be rejected.

4/ To determine drained weight, the free liquid in the can shall be poured off and the remaining contents shall be poured into a flat-bottom container. A minimum of three times the tray pack can's volume of 180°F to 190°F water shall be added to the container so as to cover the contents. The contents and water shall be gently agitated so as to remove the sauce and separate the beans. The contents shall then be poured into a U.S. Standard No. 8 sieve in a manner that will distribute the product over the sieve without breaking the beans. The sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted at approximately a 45° angle and allowed to drain for two minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. The drained weight shall be reported to the nearest 0.1 ounce.

4.5.3.3 Salt content testing. Three filled and sealed tray pack cans shall be selected at random from the lot. The tray pack cans shall be individually tested for salt content in accordance with the Official Methods of Analysis of the AOAC, method 935.47, except that preparation of the samples shall be as follows: The unopened tray pack cans shall be gently warmed in a water bath to melt fat adhering to the inside of the cans. The cans shall be opened and the entire contents of each can shall be separately blended in a Waring Blendor or equivalent. The test results shall be reported to the nearest 0.1 percent. Any result failing to conform to the salt requirement in 3.6 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.3.4 Commercial sterility testing. The sample size shall be one filled, sealed, and thermoprocessed tray pack can selected from each retort batch in the lot. The sample cans shall be tested for sterility by incubating the cans at 95°F ± 5°F for ten days. Any evidence of can swelling shall be considered a critical defect and shall be cause for rejection of the lot.

4.5.4 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the U.S. Standards for Condition of Food Containers, except that the inspection for labeling shall be in accordance with 4.5.4.1. In addition, scratches, scuffs, or abrasions that occur on the outside coating as a result of the filling, sealing, and thermoprocessing of the tray pack cans shall not be scored as a defect.

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4.5.4.1 Can label examination. Labels shall be examined for defects in accordance with MIL-L-1497 (see 5.4), except, for self-adhering labels, the following additional defects shall apply:

Major: Label torn or scratched so as to obliterate any of the markings

Minor: Air bubbles under label
Label not properly adhered to can; for example, label raised or peeled back from edges or corners

4.5.4.2 Label adhesive examination. When self-adhering labels are used, the adhesive shall be tested in accordance with ASTM D 3330.

4.5.5 Can closure examination. Can closure shall be examined visually and by teardowns in accordance with the can manufacturer's guidelines/ requirements and 21 CFR, Part 113, Subpart D, or 9 CFR, Part 318, Subpart G, as applicable. Any nonconformance based on observation of can seam teardowns or on record of can seam teardowns shall be classified as a major defect and shall be cause for rejection of any involved product.

4.5.6 Vacuum examination. Cans shall be allowed to cool to 70°F ± 5°F, held for at least 24 hours after sealing, and then examined for vacuum retention. To examine, lay a straight edge in the center of the lid along the length of the tray pack. Both ends of the straight edge shall touch the lid at the inside edge of the double seam. There shall be a visible gap between the straight edge and the lid for the entire distance of the label panel. Using a shorter straight edge, the same procedure shall be used across the width, in the center of the tray pack can. When examining a ribbed lid, lay the straight edge only between the two center ribs along the length of the can. The inspection lot shall include only tray packs produced in a single shift on a single sealing machine. The sample size shall be 50 cans. Any nonconformance shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.7 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible
Reinforced with other than nonmetallic strapping or tape
Dimensions of pads not as specified
Interior packing with fiberboard liner or pads not as specified

Minor: Other required markings missing, incorrect, or illegible
Arrangement or number of cans not as specified

4.5.8 Unit load examination. Examination of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

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5. PACKAGING

5.1 Preservation. The product shall be preserved in accordance with level A.

5.1.1 Level A. One hundred and six ounces of food product shall be filled into a tray pack can conforming to MIL-C-44340 and sealed and thermoprocessed as specified in 3.4 and 3.5.

5.2 Packing. The product shall be packed in accordance with level A, B, or C as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L or HSC-L with a HSC full depth cover, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box, with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads fabricated of grade V3c fiberboard. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). Flute direction of the box liner shall be vertical. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box. Each box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament-reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1) except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern. Boxes may be stacked by interlocking and reversing each tier, or by columnar stacking with paperboard or fiberboard sheets placed between each tier. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Four cans of product, preserved as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.3 Level C packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). Flute direction of the box liner shall be vertical. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of class domestic, grade 175 fiberboard.

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5.3 Unit loading. When specified (see 6.1), the product, packed as specified in 5.2.2 or 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern. Boxes may be stacked by interlocking and reversing each tier, or by columnar stacking with paperboard or fiberboard sheets placed between each tier. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497 and with the following:

- Official establishment number (for example, EST 38) or a three-letter code identifying the establishment
- Lot number 1/
- Production shift number 1/
- Retort identification number 1/
- Retort cook number 1/

1/ The lot number shall be expressed as a four-digit Julian code. The first digit shall indicate the year of production and the next three digits shall indicate the day of the year (for example, March 19, 1992 would be coded as 2078). The Julian code shall represent the day the product was packaged and processed. Sublotting (when used) shall be represented by an alpha character immediately following the four digit Julian code. Following the four digit Julian code and the alpha character (when used), the other required code information shall be printed in the sequence as listed above.

In addition, the name of the product shall be marked, stamping is permitted, on one 1001 by 200 end of the can. The labeling shall be legible when examined as specified in 4.5.4.1 after preparation of the product in accordance with heating instructions. Paper labels are not permitted. Cans shall show the following statements:

TO HEAT IN WATER: Submerge unopened can in boiling water. Simmer gently for 40 to 45 minutes. Avoid overheating (can shows evidence of bulging).

CAUTION: Use care when opening as pressure may have been generated within the can.

TO HEAT IN OVEN: Either punch several holes in the lid of the can or open the can in the usual manner leaving the loose lid in place. Place in a 350°F oven for 35 to 40 minutes.

WARNING: Do not place unopened can in oven. This may cause the can to burst.

YIELD: Serves 18 portions of 2/3 cup each.

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As an alternate labeling method, a preprinted, self-adhering, 0.002-inch thick, clear polyester label printed with indelible black ink may be used. Self-adhering labels shall be applied after retorting. Pressure-sensitive adhesive shall require no preparation prior to application. Labels shall tack quickly and adhere without curling or breaking. The adhesive shall have a minimum adhesion of 60 ounces per inch width when examined as specified in 4.5.4.2. When self-adhering labels are used, the tray pack can shall be labeled with the Julian code and a product code prior to retorting.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, the following precautionary markings in capital letters larger than other markings shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT, OR MORE THAN TWO HIGH IN STORAGE, UNLESS PALLET RACKS ARE USED.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.4, and 6.2).
d. Provisions for approved preproduction samples (see 3.6.1 and 6.2).
e. Level of packing required (see 5.2).
f. Type and class of unit load when unit loading is required (see 5.2.1 and 5.3).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

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6.3 Subject term (key word) listing.

Bacon
Canned food
Combat field feeding
Food processing
Meat
Operational rations
Shelf stable

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Preparing activity:

Army - GL

(Project 8940-0749)

Review activities:

Army - MD, QM
Navy - MC
DLA - SS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-44242C	1993 August 16

3. DOCUMENT TITLE
PORK AND BEANS, THERMOSTABILIZED, TRAY PACK

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON/DSN
U.S. Army Natick RD&E Center	508-651-4501	256-4501
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-STP Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 NATICK/SUPERSEDING [illegible] 109/290

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